AFNet: Adaptive Fusion Network for Remote Sensing Image Semantic Segmentation

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***Abstract*— Semantic segmentation of remote sensing images plays an important role in many applications. However, a remote sensing image typically comprises a complex and heterogenous urban landscape with objects in various sizes and materials, which causes challenges to the task. In this work, a novel adaptive fusion network (AFNet) is proposed to improve the performance of very high resolution (VHR) remote sensing image segmentation. To coherently label size-varied ground objects from different categories, we design multilevel architecture with the scale-feature attention module (SFAM). By SFAM, at the location of small objects, low-level features from the shallow layers of convolutional neural network (CNN) are enhanced, whilst for large objects, high-level features from deep layers are enhanced. Thus, the features of size-varied objects could be preserved during fusing features from different levels, which helps to label size-varied objects. As for labeling the category with high intra-class difference and varied scales, the multiscale structure with a scale-layer attention module (SLAM) is utilized to learn representative features, where an adjacent score map refinement module (ACSR) is employed as the classifier. By SLAM, when fusing multiscale features, based on the interested objects scale, feature map from appropriate scale is given greater weights. With such a scale-aware strategy, the learned features can be more representative, which is helpful to distinguish objects for semantic segmentation. Besides, the performance is further improved by introducing several nonlinear layers to the ACSR. Extensive experiments conducted on two well-known public high-resolution remote sensing image data sets show the effectiveness of our proposed model. Code and predictions are available at https://github.com/athauna/AFNet/.**

***Index Terms*— Attention mechanism, convolutional neural network (CNN), semantic segmentation.**

I. Introduction

SEMANTIC segmentation aims to assign each pixel in the image with a predefined semantic label. In the domain of remote sensing, automatically labeling of remote sensing imagery is essential for a wide range of applications. For instance, land cover which is fundamental to the physical environment [1], and change detection which analyzes potential changes between multidate images [2], both require image segmentation results. Owing to the widespread applications, the segmentation of very high resolution (VHR) images has been one of the most vibrant fields of research in the remote sensing community. This article considers the segmentation algorithm of VHR images in the urban area.

A considerable amount of literature has been published on the VHR image segmentation task. The early research focused on learning features from local portions of the image and then designed a supervised classifier to identify the learned features to label space [3]–[5] or into bag-of-words [6]. Recently, the convolutional neural network (CNN) achieves state-of-the-art (SOTA) performance in many computer vision tasks. Motivated by its remarkable improvements compared with traditional machine learning approaches, scholars attempt to apply it to the remote sensing society. From scene classification [7]–[9] to semantic segmentation. The approaches on segmentation vary from the first patch-based [10], [11] methods to a large variety of fully convolutional networks (FCNs) [12]–[14] and deeplabs [15], [16]. Despite the dramatic performance of current CNN-based methods, however, there are still some deficiencies.

The first deficiency is simultaneously labeling size-varied ground objects. On VHR images, there are small or threadlike objects, as well as large objects which should be segmented simultaneously. The hierarchical structure of CNN could solve the problem in theory, since the semantic information of small objects exists in the shallow layer while the semantic information of large objects is in the deep layer. However, previous studies either do not consider shallow layer or merely merge semantic information from the shallow and deep layer. It means that the semantic information of the small objects may disappear or be covered by the semantic information of the large objects when fusing multilevel features. Adaptively select shallow or deep features based on image content could be a possible solution to this problem. The attention mechanism, by which we can selectively focus on parts of the image and then gradually combine the information from a different area, can theoretically solve this problem. To this end, we propose scale-feature attention module (SFAM) to address this problem. By SFAM, when fusing feature maps from multilevels, for each point on the feature map, if it represents a small or threadlike object on the original image, then low-level features from the shallow layer will be given greater weight, otherwise, high-level features from the deep layer will be considered more important. Through this adaptively

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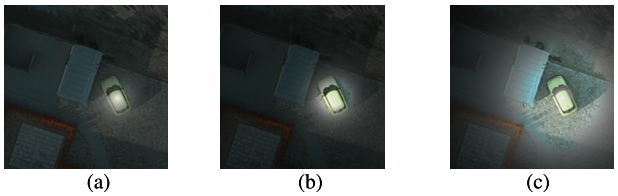


Fig. 1. Brighter area indicates the receptive field. From left to right is feature maps with small, moderate, and large receptive fields, respectively. For the recognition of the car, on large scale, the semantic information of roof can be ambiguous. While on the small scale, the semantic information of the car is insufficient. (a) Small scale. (b) Suitable scale. (c) Large scale.

weighting strategy, features of size-varied objects could be learned jointly, which could help to address the labeling problem.

The second deficiency is the segmentation of easily confused ground objects. Generally, ground objects usually exhibit higher intraclass variances and lower interclass variances. For instance, roofs can be made of different materials, while roofs and roads can be composed of the same material. Previous studies utilize multiscale structures to extract feature maps with large receptive fields to recognize those challenging objects. As effective as those approaches are, however, using fixed dilated rates or strides during multiscale feature extraction, ignoring the scale of challenging objects might lead to an inappropriate receptive field, which means the receptive field mismatch the scale of objects, and this mismatch has an impact on accuracy. Fig. 1 shows an example of the inappropriate. In essence, the suitable receptive field is depended on the scale of the object, so attention mechanism is also helpful to the mismatch problem. Thus, scale-layer attention module (SLAM) is proposed to address this derived mismatch receptive field problem. Several attention maps are generated by SLAM. Based on the scale of the target object, the feature map from the corresponding scale will be enhanced by the generated attention map when fusing the multiscale features. This scale-aware fusing strategy helps to learn more representative features of easily confused objects, thus completing the semantic segmentation of those objects. Besides, apart from learning feature representation, improving the ability of the classifier can also be beneficial to the segmentation task. So, another module named adjacent score map refinement module (ACSR) serves as a nonlinear classifier to further improve the accuracy.

In contrast to previous work that pays attention to design elaborate modules to extract more contextual information or refine the boundaries, our research explores the following four aspects.

1) Combining the advantages of multiscale and multilevel network structures.
2) The influence of the fusion strategy of feature maps from different levels.
3) The effects of the fusion strategy of feature maps at different scales with various receptive fields.
4) The classifier utilized in CNN. To our best knowledge, the second and third aspects have not yet explicitly been addressed in recently published articles.

Our contributions can be summarized as follows:

1) A novel framework named adaptive fusion network (AFNet) is proposed for VHR remote sensing image segmentation. With the two proposed attention-based modules, i.e., the scale-feature attention and the SLAMs, AFNet can adaptively focus on size-varied objects on multilevel feature maps and search for the most appropriate scale for challenge objects in multiscale structures.
2) The SFAM is introduced to help the size-varied objects labeling in VHR image semantic segmentation. The module can adaptively enhance feature points from low-level or high level feature maps based on the categories of the points on the original image.
3) To segment challenging objects more accurately, a SLAM is introduced. The feature map with an appropriate receptive field that matches the scale of the object is enhanced by this module, thus making the learned features more representative for the segmentation of the challenging objects. In addition, by introducing several nonlinear layers, ACSR is utilized as a more competitive classifier to further boost the performance.

The rest of this article is organized as follows. Section II introduces some literature of image segmentation using deep learning, both in the domain of remote sensing and computer vision. Section III describes our proposed model in detail. Section IV shows our experimental results compared with other SOTA methods. Discussions are presented in Section V, while conclusions are illustrated in Section VI.

II. Related Work

A. Segmentation in Computer Vision

Recently, introducing attention mechanism to CNN has become a rising tendency in computer vision. To our knowledge, the first approach is based on channel attention. EncNet [17] utilizes a context encoding module to capture the encoded semantics and predict scaling factors that are conditional on these encoded semantics. Yu *et al.* [18] developed channel attention block to combine features from adjacent stages. Specifically, high level features provide semantics to guide the selection of low level features in the channel direction. Li *et al.* [19] utilized semantic information from the high level as an attention guide to strengthen the weights of the parts with object information.

However, channel attention-based models do not take spatial relationships into consideration. To remedy this, Mou *et al.* [20] designed a spatial relation module to model the spatial relation between the target feature points and other feature points (also called long-range information). Nevertheless, the amount of computation to model the spatial relation for each feature point is too large. To reduce calculation, Li *et al.* [21] utilized expectation maximum (EM) algorithm to optimize the attention module to heavily reduce calculation. Likewise, Zhu *et al.* [22] designed asymmetric pyramid nonlocal module to reduce the computation and memory consumption.

Nevertheless, these spatial relation modules are designed to capture long-range information. They are not necessary for

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small patches scratched from VHR images. Different from theirs, we utilize a multiscale structure to capture long-range information.

B. Remote Sensing Image Semantic Segmentation

There is plenty of literature on remote sensing image segmentation. Previous studies are divided into the following categories, patch-based methods, multilevel methods, and multiscale methods. Only the latter two types of methods are introduced here.

Multiscale methods are proposed to overcome the obstacle that ground objects have high intra-class variances and low interclass variances. For instance, Liu *et al.* [23] utilized dilated convolution [15] with different dilated sizes on the last layer of the encoder part to capture contextual. Yue *et al.* [16] utilized spatial pyramid pooling [24] layers on feature maps of the last convolutional layer. However, these multiscale methods fail to refine the object boundaries.

Multilevel methods can recover the resolutions of feature maps, refine the relatively coarser prediction and precisely segment the boundaries of fine-structure objects. In every link of the upsampling procedures, well-designed modules are used to refine the boundaries of fine-structured objects. Mou and Zhu [13] utilized feature maps from four levels to segment vehicles on VHR images. Mi and Chen [25] utilized the downsampled and image-level feature maps to refine the object boundaries. Li *et al.* [26] designed an adaptive fusion module for the fusion of feature maps from three levels of the backbone network. However, its ability to identify challenging objects is worse than multiscale methods.

Nevertheless, no matter multiscale or multilevel methods, they all fuse feature maps with different receptive fields in a premature way. Considering a single feature map with a relatively large receptive field, though there is enough contextual information, the feature of the target object might be overlaid by other unrelated objects. This could be worse if the target object is relatively small (e.g., the semantic information of vehicles on low level feature maps is richer than on high level feature maps). Meanwhile, feature maps which possess small receptive fields, the learned features may lack contextual information, which is harmful for segmentation. If fusing these features without adjustment, the ambiguity may be introduced to the segmentation of challenging objects. This phenomenon is named as receptive fields imbalance proposed in [27].

C. Classifier in Remote Sensing Image Semantic Segmentation

Usually, softmax is utilized as a nonparameter classifier in CNNs. Here, we regard the module that generates the input of softmax as a trainable classifier. In theory, CNN can learn feature representations and parameters of trainable classifier simultaneously. However, current research studies pay more attention to the design of CNN. In fact, most multilevel and multiscale structures only utilize point-wise convolution as a classifier. On the contrary, because of the limited power of learning feature representations, patch-based methods usually utilize more competitive feature representations obtained by nonlinear layer. To be concrete, Zhang *et al.* [10] utilized a multilayer perceptron (MLP) and CNN as two classifiers, and the prediction is obtained by a rule-based decision fusion approach based on the prediction obtained by MLP and CNN. Audebert *et al.* [28] used three convolutions with different kernel sizes as a classifier to acquire multiscale predictions. Maggiori *et al.* [29] designed an independent neural network as a classifier to classify the combined features. Moreover, Liu *et al.* [30] used feature fusion learning to classify the fused spectral and spatial features. Owning to the complexity of challenging ground objects, apart from the design of the feature extraction part, our work focuses on the comprehensive use of score maps and feature maps from different levels and replaces the linear classifier with a nonlinear multilayer classifier.

III. Methodology

This section is organized as follows, first, in Section III-A, an overview of the proposed model for VHR remote sensing images is illustrated. Then the three main modules, SFAM, SLAM, and ACSR are introduced below. For feature maps from atrous spatial pyramid pooling (ASPP) [31], SLAM is utilized to balance the receptive field during the multiscale feature fusion. For the labeling of size-varied objects, fusing feature maps from different levels is necessary, and SFAM is designed to model the correspondences of pixels on different levels of feature maps. After feature extraction, ACSR works as a classifier to produce final predictions.

A. Pipelines of Proposed Model

For the segmentation of the challenging objects, the multiscale structure is essential, since it generates feature maps with relatively large receptive fields, which are beneficial to the segmentation of the challenging objects. For the segmentation of the size-varied objects, feature maps from multilevels could be helpful, since feature from the shallow layer of CNN can provide semantic information of small or threadlike objects and feature from deep layer contains semantic information of relatively large object. Combining the advantages of multiscale and multilevel structures, we propose a hybrid model to solve the problem of segmentation of ground objects on VHR remote sensing images. SFAM and SLAM are designed to help with labeling size-varied objects and easily confused objects, separately. The pipeline of the proposed model is shown in Fig. 2. As the figure shows, the network can be divided into three parts, the encoder part, the decoder part, and the classifier part.

In the encoder part, according to [32], pretrained ResNet [33] is used as the backbone network to extract features. Dilated convolutions with different rates and image pooling are adopted to capture long-range information. After each convolution, SLAM is applied to all the feature maps, thereby completing the feature extraction task of the first stage. In the decoder part, we use three different size feature maps from the encoder for fusion. The fusing process is gradually completed by the three SFAMs. As for the last procedure, we use ACSR module composed of several point-wise (the size of the convolution kernel is 1) convolutions to generate score map. However, the highest level score map is generated in the

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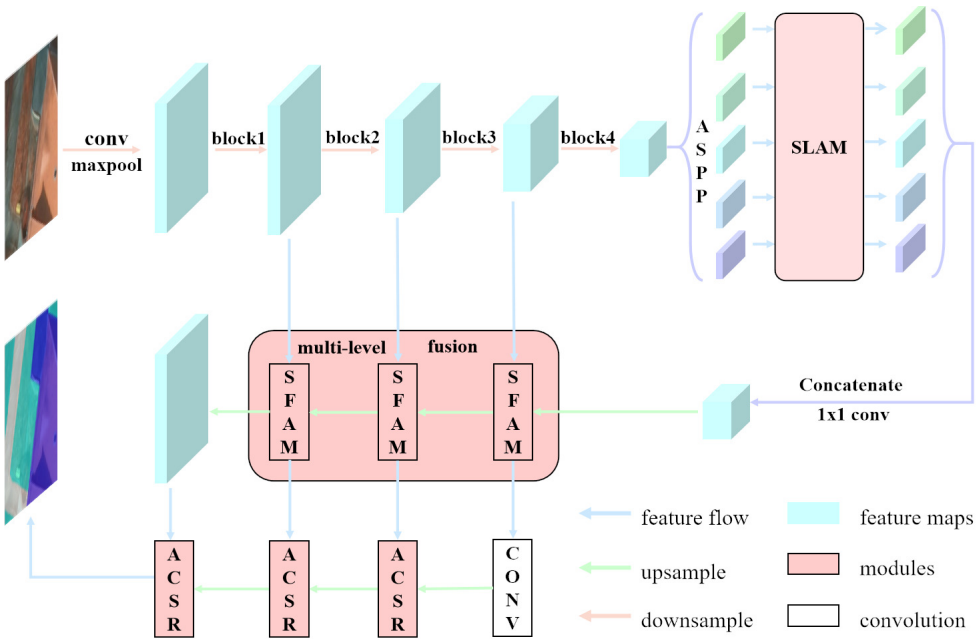


Fig. 2. Pipeline of proposed networks. It is composed of several SFAMs, SLAMs, and ACSR blocks. Cubes with colors mean operations such as convolution or designed module.

traditional way, since there is no higher level score map as the input of ACSR. As is commonly utilized in segmentation, cross entropy is used as the loss function.

B. Scale-Feature Attention Module

The conventional fusion strategy achieved by deconvolution does not model the correspondences of pixels on hierarchical feature maps, which might account for the challenge that exists in labeling size-varied objects. Inspired by the nonlocal network [34], we improve the fusion process by a module named SFAM.

The module is explained as follows, suppose that x_l is a 1-D vector from the channel direction of the low level feature map while x_h is from high level feature map. x_l has N channels. When without special instructions, x_l and x_h denote pixels in the same position on low level feature map and the resized high level feature map. Then we use the similar formula in [34] to represent the fusion process of x_l and x_h

$$x_o = \frac{1}{C(x)} f(x_l, x_h) \cdot g(x_l). \tag{1}$$

In (1), x_o is the output of this fusion process. f is the pairwise function which calculates the relationship between x_l and x_h. $C(x)$ is the normalization factor, and g is a unary function which computes a representation of the input vector x_l. As the equation suggests, for each element in x_l, the information from x_h is used to produce response, and this response can be regarded as a weight scalar for $g(x_l)$.

To convert the formula into a computable neural network module, the above function f, g, and the normalized factor $C(x)$ need to be instantiated. The Gaussian function is the natural choice of f, however, considering the implementation, embedded Gaussian is chosen here, which means the pairwise relationship is computed in an embedded space

$$f(x_h, x_l) = e^{\theta(x_l) \cdot \phi(x_h)} \tag{2}$$

where θ and ϕ are the two embeddings which can be implemented as simple point-wise convolutions ($\theta(x_l) = W_\theta x_l$). Notice that the multiplication here is hadamard product of vector $\theta(x_l)$ and $\phi(x_h)$. Although x_l and x_h do not necessarily have the same number of channels, $\theta(x_l)$ and $\phi(x_h)$ can have the same number (the number can be controlled by W, usually predefined as $\bar{N}$) of channels indeed. This is another reason why we use convolution as embedding. As for function g, here still in the form of point-wise convolution. To keep the channel number consistent with the output of function f, the vector output by function g also has $\bar{N}$ channels. As for normalized factor $C(x)$, here $C(x) = \sum_{i=1}^{\bar{N}} f(x_l, x_h)$. Then (1) can be rewritten as

$$\begin{aligned} x_o &= \frac{e^{\theta(x_l) \cdot \phi(x_h)} \cdot g(x_l)}{\sum_{i=1}^{\bar{N}} e^{\theta(x_l) \cdot \phi(x_h)}} \\ &= \text{softmax}(\theta(x_l) \cdot \phi(x_h)) \cdot g(x_l). \end{aligned} \tag{3}$$

Moreover, to avoid gradient vanishing, here residual connection [33] is used. The output is concatenated with the representation of x_l, thus the final output of this block is $\text{Out} = x_o + x_l$.

For the convenience of understanding, the above argumentation is vector-based, but as hadamard product, point-wise convolution and softmax function work same on vector and matrix, which means that our module can also take matrix as input. The differences between our SFAM and [34] are in three aspects. First, the two modules share different motivation. Wang *et al.* [34] is designed to capture long range or global information, while SFAM is more likely to focus on local region. Second, the inputs of the two modules are not the

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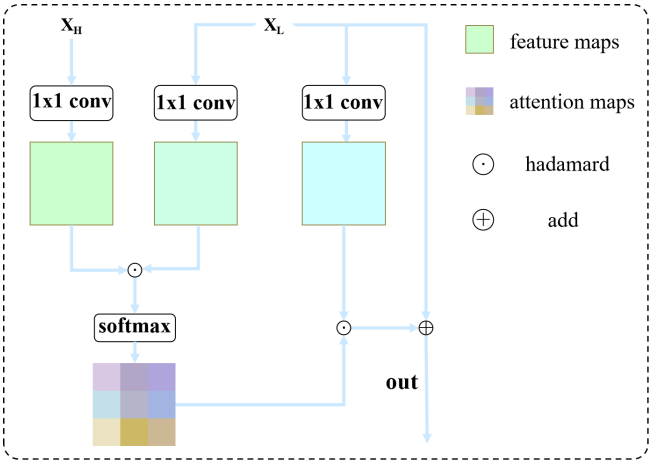


Fig. 3. Our proposed SFAM module to fuse feature maps from different levels. Feature maps from adjacent levels are put into this module to generate the attention map. The generated attention map is multiplied by low level feature maps. Still, the residual connection is utilized here to avoid gradient vanishing.

same. Feature maps from different levels are input into SFAM, but [34] takes the single level feature map as input. Third, the calculation of SFAM and [34] network is different. Hadamard product rather than matrix multiplication in [34], is used to calculate the attention map. Moreover, suppose there are N pixels on the feature map, for each pixel, SFAM only needs a pair of pixels to calculate the attention value, but [34] needs N pairs of pixels. From this perspective, our module is more computationally efficient than [34], which also explains why it can be used three times in the network. Fig. 3 shows the above process in the form of matrix.

C. Scale-Layer Attention Module

Since remote sensing typically comprises a complex and heterogenous landscape, the challenging object could not be recognized well by multilevel networks. The main reason is the lack of sufficient contextual information, so the ASPP module is introduced to solve this problem, and it has been proven to be effective in the segmentation of VHR remote sensing images [23], [25]. However, when extracting contextual information, the dilated rates utilized in ASPP are fixed, ignoring the scale of challenging objects. This may cause the receptive field of the extracted feature map does not match the scale of the challenging object. A module named SLAM is designed to generate attention map, and during fusing the feature maps from ASPP module, the attention maps are utilized to adaptively weight the multiscale feature maps. Those feature maps have the same spatial resolution but the receptive fields of the pixels are different. According to the scale of challenging object, feature map with matched receptive field will be enhanced while others will be restrained. Our approach is distinct in that 1) the module focuses on the pixel itself to generate attention value, rather than a large region around it or its correlations with global pixels and 2) this module could be parameterized, which means it is differentiable and it can be trained in an end-to-end manner. By SLAM, we explicitly model the discrimination of each

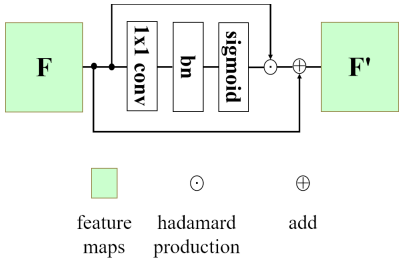


Fig. 4. Our proposed SLAM module to weight feature maps from different scales. Depth-wise convolution and sigmoid function are utilized to generate weight map. In addition, we use residual connection. That means the output of this module is the sum of weighted feature map and original feature map.

pixel from the same level, adaptively fuse feature maps with different receptive fields. SFAM allows the model to make comprehensive use of the multiscale structures and attention mechanism, while remedying the receptive fields imbalance.

First, we will introduce the compositions of this SLAM module. As illustrated in Fig. 4, the output of the module can be denoted as

$$F^o = F_w \cdot F + F \tag{4}$$

where F denotes the input of SLAM, F_w is the attention map, and F^o is the output feature map. The key point is to generate an attention map that is only related to the target pixel itself. The process of generating F_w can be denoted as

$$F_w = \text{Sigmoid}[\text{BN}[k \otimes F]] \tag{5}$$

where k means parameter of point-wise convolution, $\otimes$ indicates convolution operation. BN is batch normalization function [35]. This equation reveals that F_w is differentiable. Here, we assume that the module can dependently generate attention in channel direction, so $Sigmoid$ function is selected to calculate attention values in channel direction rather than $softmax$ which takes other values in channel direction into computation.

Then the input feature map F is multiplied with the attention map. Since the initialization of F_w depends on the initialization of the convolution kernel, there might exist the gradient vanishing problem. So, here residual connection [33] is utilized. The worst situation is that F_w is close to zero at the beginning, but by this residual connection, the whole net can be trained normally to update F_w and other parameters automatically. For each feature map from the ASPP module, the SLAM is applied on. In this way, all feature maps from the ASPP module have its corresponding weights both in spatial and channel direction.

The difference between channel attention module and our attention modules is illustrated below. Typically, channel attention is achieved by channel-wise multiplication, which can be denoted as

$$F_k^o = \alpha_k \cdot F_k \tag{6}$$

where F_k means feature map from k_{th} channel, α_k is a scalar. As the equation indicates, feature map from k_{th} channel is multiplied with a fixed parameter, ignoring the diversities of pixels from the spatial direction. Similar logic applies to other cases where channel attention and spatial attention are

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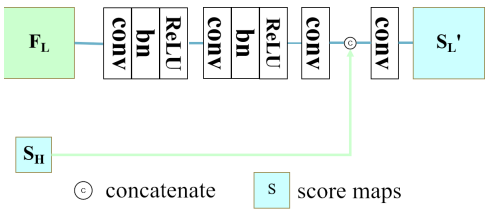


Fig. 5. Structure of our proposed adjacent-connected score map refine block. It takes the score map from the previous stage and feature maps from the same stage as inputs and outputs the refined feature map. Same as in SFAM, the blue line means feature map, and the green line means resized feature map.

implemented separately. However, our location-wise attention works as follows:

$$f^o(i, j, k) = w(i, j, k) \cdot f(i, j, k) \tag{7}$$

where (i, j, k) means the pixel is in the i_{th} row, j_{th} column in the k_{th} channel of the given feature map, w is the corresponding weight value. This means we not only achieve channel attention but also take the diversity of pixels from the same spatial into consideration.

D. Adjacent-Connected Score Map Refine

When it comes to classification tasks, after flatten or global pooling, a fully connected layer works as the classifier. However, we observe that in most current segmentation tasks, point-wise convolution is substituted for the fully connected layer. Due to the linear property of point-wise convolution, the classification ability of point-wise convolution is weaker than that of a fully connected network. Another phenomenon worth to be mentioned is the comprehensive use of score maps from different stages. Previous work pays little attention to score maps from different levels or just add them together, ignoring the comprehensive use of score maps. That is the motivation that we propose the adjacent connection score map refine module.

Different from [10], our proposed ACSR is a pixel-based classifier which is designed to make comprehensive use of score maps. By introducing several nonlinear layers, the capacity of the classifier is powerful than the naive classifier, which is composed of the point-wise convolution layer. To make comprehensive use of score maps, we choose to concatenate score maps from adjacent levels and then apply point-wise convolution on them. More details of ACSR module are shown in Fig. 5. The input feature map is first input to the assembled point-wise convolution to produce preliminary score map. For a given feature map, to determine the category of each pixel, the vector from channel direction should be utilized, let $\mathbf{v}$ denote the vector. For category c, the conventional point-wise convolution works as

$$S_c = \mathbf{w}^T \cdot \mathbf{v} \tag{8}$$

where S_c represents the score that the pixel belongs to category c. After softmax, it is transformed to probability. $\mathbf{w}$ is the parameter of single point-wise convolution. n is the length of the vector. But in our approach, the adapted classifier for category c is quite different. First, we define function f as follows:

$$f(\mathbf{v}) = \text{ReLU}[\text{BN}[\mathbf{W}^T \cdot \mathbf{v}]]. \tag{9}$$

TABLE I

DETAILED INFORMATION OF SETTING ON EXPERIMENTAL DATA SETS. VAL IS THE ABBREVIATION OF VALIDATION SUBSETS, GSD MEANS GROUND SAMPLING DISTANCE, RESPECTIVELY

	Train	Val	Test	Category	GSD
Potsdam	20	4	14	6	5 *cm*
Vaihingen	12	4	17	6	9 *cm*

Then our classifier can be denoted as

$$S_c = \mathbf{w}^T \cdot f(f(\mathbf{v})). \tag{10}$$

The combination of several convolution layers with rectified linear unit (ReLU) introduces nonlinear property. After that, score maps from $(k-1)_{\text{th}}$ level and k_{th} level are concatenated together, then point-wise convolution is applied to produce the score map of k_{th} level.

IV. EXPERIMENTS

We evaluate our proposed model on Potsdam and Vaihingen data set provided by International Journal of Photogrammetry and Remote Sensing (ISPRS) [36] 2-D semantic labeling contest. In this part, we first introduce the data set, and then training details and evaluation metrics. After that, we will compare our results with SOTA methods. We perform comprehensive ablations to validate each component of our model in Section V.

A. Data Set Description

1) Potsdam Data Set: There are six categories in the data set, which are impervious surface, building, low vegetation, tree, car, and background. The Potsdam data set has a total of 38 tiles, of which 14 tiles are reserved as the test set. All tiles have red, blue, green, near-infrared bands, normalized digital surface model (nDSM) data, and DSM data. The size of each tile is 6000×6000 pixels. From the training set, we randomly pick four tiles as validation data. For each tile, only near-infrared, red, and green bands are chosen. When training, each training tiles are cropped randomly to 4000 patches with a size of 256×256 pixels. As for validation and testing, we crop patches in a nonoverlapping manner.

2) Vaihingen Data Set: Released by the same organizers, this data set has similar properties. It also contains six categories, four optical bands, DSM, and nDSM data. However, compared with the Potsdam data set, the ground sample distance (GSD) is lower. It contains 33 images of around 2500×2000 pixels per image. 17 of them are preserved as test set while 16 of them serves as a training set. In addition, owing to the relatively small size of titles, for a given image, we randomly crop it to produce image patches with an image size of 256×256 pixels as training data.

Here, all ablation studies report labeling performance on the validation set we split. The final model on test data is retrained on the whole training set. Table I summarizes the detailed information of data sets illustrated above, and Fig. 6 shows some examples and ground truth of these data sets. As Fig. 6 shows, for the Potsdam data set, there are confusing objects which introduce an extra challenge for the labeling task. Though the number of training images seems enormous, for deep learning-based method, the number of tiles is still not enough.

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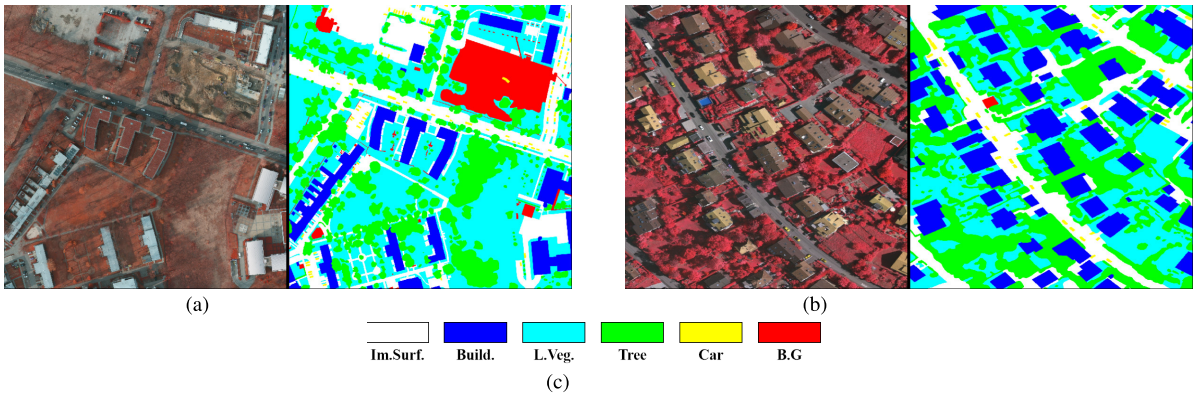


Fig. 6. Image sample and corresponding ground truth of Potsdam and Vaihingen challenge. The labels of Potsdam and Vaihingen challenge include six categories: impervious surface (white), building (blue), low vegetation (cyan), tree (green), car (yellow), and background (red). (a) Potsdam challenge. (b) Vaihingen challenge. (c) Legend.

B. Training Details

1) Avoid Overfitting: Considering that the scale of data sets is relatively small, to avoid overfitting, transfer learning and data augmentation techniques are adopted here. According to [32], ResNet which has been pretrained on ImageNet [37] is chosen to be our backbone network for both data sets during all experiments. For each image, random flip, randomly rotate and multiscale input are utilized as data augmentation.

2) Configuration of Hyper Parameters: The experiments are running with Tensorflow [38] tool. As for hyperparameters, for both data sets, we adopt stochastic gradient descent (SGD) with momentum 0.9 as an optimizer, set the initial learning rate to 0.005. Similar to [39], here we employ *poly* learning rate policy where initial learning rate is multiplied by $(1-(\text{iter}/\text{max_iter}))^{\text{power}}$ with power = 0.9 to change learning rate. The network is first trained for 300 000 iterations and the learning rate is divided by 10 for another 150 000 iterations.

3) Configuration of Network: To avoid coarse prediction, the *output stride* of ResNet is set to 16. As for the dilated convolution rate in ASPP module, we set it to (6, 12, 18), respectively. Inspired by [31], [40], we also use the multigrid method in the backbone, and the grid value is set to (1, 2, 4). Multigrid method is initially used to solve elliptic equations. The error of numerical solution can be expanded into the form of series, grids of different scales can filter out errors in different series. This idea is transformed in CNN [41], [42]. The multigrid method employs a hierarchy of grids of different sizes to adjust the rates of dilated convolutions. The number of grids depends on the number of repetitive units in the last block of ResNet. The final dilated rates of the three convolutions are the multiplications of the corresponding unit rates and the constant determined by the predefined output stride. In our case, when applied multigrid methods, the dilated rates of three convolutions will be 2, 4, and 8, respectively. Limited by GPU memories, we set *batch size* to 12.

C. Evaluate Metrics

To assess quantitative performance, four common and widely accepted metrics are utilized here. They are F_1 score, overall accuracy (OA), average accuracy (AA), and intersection over union (IoU). Before the definitions of the four metrics, related initialisms will be introduced. To be concrete, *tp, fp, fn*, and *tn* are the representations of true positives, false positives, false negatives, and true negatives. Thus, *precision* (P) and *recall* (R) can be calculated as

$$\mathrm{P} = \frac{\mathrm{tp}}{\mathrm{tp}+\mathrm{fp}}, \quad \mathrm{R} = \frac{\mathrm{tp}}{\mathrm{tp}+\mathrm{fn}}. \tag{11}$$

So, OA can be defined as

$$\mathrm{OA} = \frac{\mathrm{tp}}{\mathrm{tp}+\mathrm{fp}+\mathrm{fn}+\mathrm{tn}}. \tag{12}$$

F_1 score is defined as

$$\mathrm{F}_1 = 2\frac{\mathrm{P}\cdot\mathrm{R}}{\mathrm{P}+\mathrm{R}} \tag{13}$$

IoU is defined as follows:

$$\mathrm{IoU} = \frac{\mathbb{S}_p \cap \mathbb{S}_{\mathrm{gt}}}{\mathbb{S}_p \cup \mathbb{S}_{\mathrm{gt}}} \tag{14}$$

where $\cap$ means the intersection, $\cup$ denotes the union of two sets. $\mathbb{S}_p$ means the set of prediction while $\mathbb{S}_{\mathrm{gt}}$ means the set of ground truth images. AA is defined as the average of *recall* of each category, which indicates the ratio of correctly classified pixels. To evaluate the performance, we calculate the aforementioned metrics on each data set. When testing on Potsdam and Vaihingen data set, the groundtruth whose boundaries of objects have been eroded by 3-pixel radius is recommended by the organizers.

D. Results on Potsdam

When evaluating performance on the test set, the size of image patches could be larger since we only predict one patch per iteration. According to the ablation study below, for each tile, we crop patches with size of a 600 × 600 pixels with half overlap. The final result is a vote of multiple prediction methods, which includes the flip, rotate 90°, 180°, and 270° and the original one. Brief introductions of some compared online methods will be introduced first. Afterward, the experimental results will be presented.

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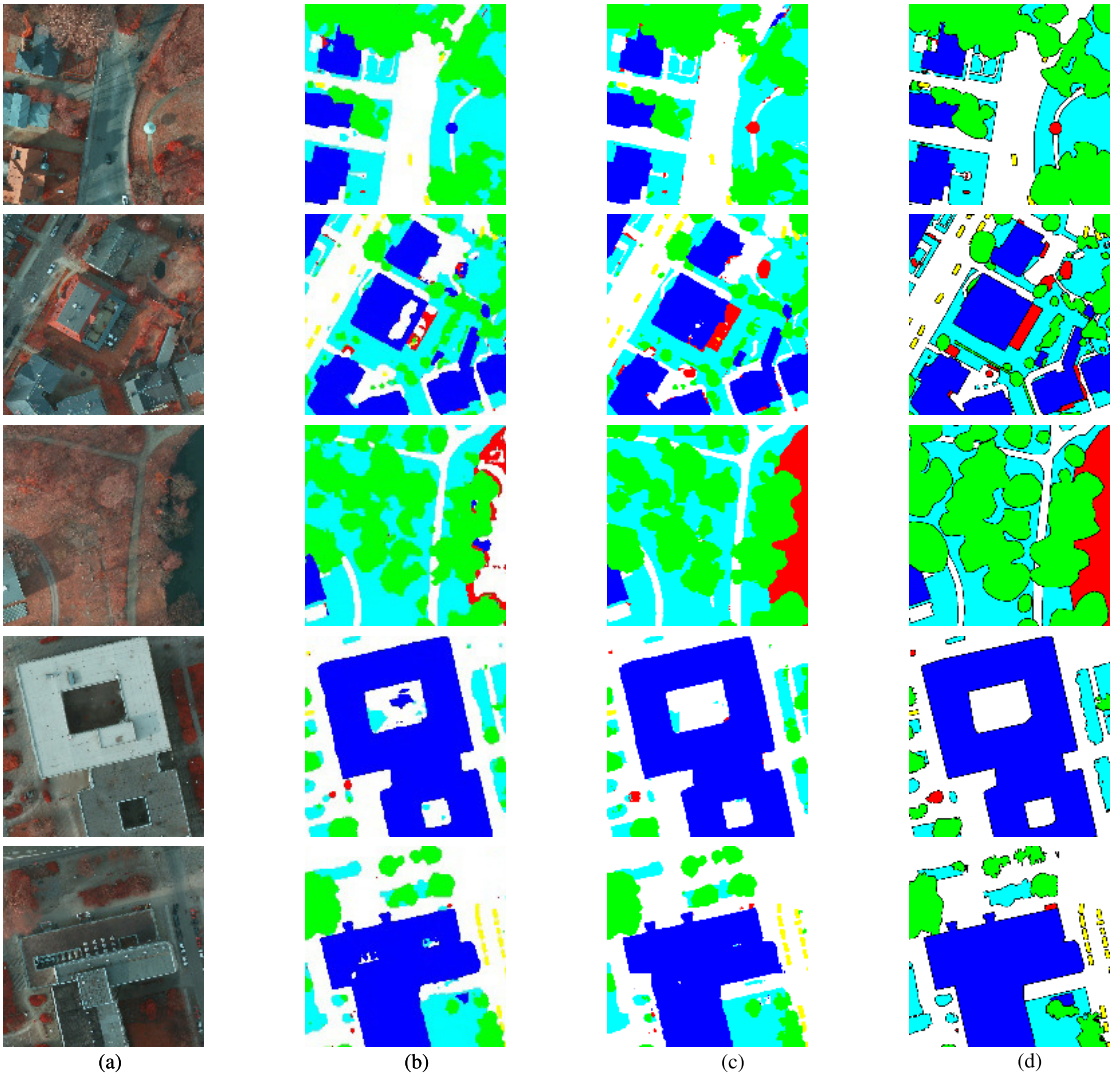


Fig. 7. Partial visualizations of results on the Potsdam test data set. From left to right are raw images, predictions by ReSegNet and our AFNet, and the ground truth labels, respectively. The colors on ground truth are, white: impervious surface, yellow: car, red: background, green: tree, cyan: low vegetation, blue: building. (a) Image. (b) ReSegNet. (c) Ours. (d) G.T.

1) TreeUnet: This model is proposed by Yue *et al.* [43] in 2019. The TreeUnet consists of three main parts, Deep-Unet for segmentation, Tree-CNN block, and concatenating connections. Tree-CNN block can be adaptively constructed via calculating the confusion matrix, building the undirected graph, and iterating the TreeCutting operation. Compared with TreeUnet, we both utilize a multilevel structure for feature extraction. However, in their work, the Tree-CNN block is utilized to transform feature maps to score maps, while in our work, we utilize ACSR which does not need iterations to update the structure. In general, after training, they need extra steps to update the Tree-CNN block and retrain the network while our method can be trained end-to-end.

2) ScasNet: This model is proposed by Liu *et al.* [23] in 2018. We both adopt Deeplabv3 as feature extraction. However, in their work, the global to local strategy is utilized to fuse feature maps from the ASPP module, and the fusion of multilevel feature maps is based on residual correction and skip connection. Different from theirs, SLAM is utilized to fuse feature maps from the ASPP module, and SFAM is utilized to fuse multilevel feature maps. On the whole, we both try to change the style of multiscale and multilevel feature fusion but achieve in different manners.

3) ReSegNet: This model is proposed by Sun *et al.* [27] in 2019. In their work, the imbalance of receptive fields is named as structural stereotype which may lead to incorrect inference on the edge area of patches. Based on SegNet [44], the residual connection is introduced as a bypass path, with several repeating units, the ReSegNet is formed. We both adopt a multilevel structure and face the challenge of receptive field imbalance. However, they try to solve it by random sampling and ensemble inference, while we do not use a predefined mask in the inference phase, but adaptively learn the mask in the training phase, with proposed SLAM and SFAM.

Table II shows the quantitative comparison with current methods which have been reported in published articles. As Table II indicates, all the methods show promising results, and our model achieved an OA of 92.2%, exceeding the

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TABLE II

QUANTITATIVE COMPARISON (%) WITH THE SOTA DEEP MODELS ON ISPRS POTSDAM CHALLENGE TEST SET, WHERE THE VALUES IN BOLD ARE THE BEST AND THE VALUES UNDERLINED ARE THE SECOND BEST. ALL VALUES ARE IN PERCENTAGES

Method	Im.Surf.	Build.	L.Veg.	Tree	Car	OA	# Para $\times 10^6$
UZ [45]	89.3	95.4	81.8	80.5	86.5	85.8	2.5
KLab_3 [14]	89.3	92.0	83.5	83.8	92.0	86.4	44
UFMG_4 [46]	90.8	95.6	84.4	84.3	92.4	87.9	2
RIT_L7 [47]	91.2	94.6	85.1	85.1	92.8	88.4	134
CVEO [48]	91.2	94.5	86.4	87.4	95.4	89.0	-
V-FuseNet [49]	92.7	96.3	87.3	88.5	95.4	90.6	-
TreeUnet [43]	93.1	97.3	86.6	87.1	95.8	90.7	-
ScasNet [23]	93.3	97.0	87.7	88.4	96.2	91.1	-
ReSegNet [27]	93.6	**97.6**	88.5	88.8	94.6	91.6	-
SDNF [25]	93.5	96.3	**89.8**	**92.7**	**96.7**	92.1	60
ours	**94.2**	97.2	89.2	89.4	95.1	**92.2**	63.4

TABLE III

QUANTITATIVE COMPARISON (%) WITH THE SOTA DEEP MODELS ON ISPRS VAIHINGEN CHALLENGE TEST SET, WHERE THE VALUES IN BOLD ARE THE BEST AND THE VALUES UNDERLINED ARE THE SECOND BEST. ALL VALUES ARE IN PERCENTAGES

Method	Im.Surf.	Build.	L.Veg.	Tree	Car	OA	# Para $\times 10^6$
UZ [45]	89.2	92.5	81.6	86.9	57.3	87.3	2.5
RIT_L7 [47]	89.6	92.2	81.6	88.6	76.0	87.8	134
ADL [50]	89.5	93.2	82.3	88.2	63.3	88.0	0.5
CVEO [48]	90.5	92.4	81.7	88.5	79.4	88.3	-
ReuseNet [51]	90.1	93.5	82.1	88.3	77.1	88.4	-
UFMG_4 [46]	91.1	94.5	82.9	88.8	81.3	89.4	2
INR [29]	91.1	94.7	83.4	89.3	71.2	89.5	4
MMDN [28]	91.0	94.5	84.4	89.9	77.8	89.8	-
V-FuseNet [49]	92.0	94.4	84.5	89.9	86.3	90.0	-
TreeUnet [43]	92.5	94.9	83.6	89.6	85.9	90.4	-
DLR_10 [52]	92.4	95.2	83.9	89.9	81.2	90.3	-
ScasNet [23]	93.2	96.0	84.7	89.9	86.7	91.1	-
ResegNet [27]	93.3	96.1	86.4	90.8	74.6	91.6	-
SDNF [25]	**93.6**	**96.2**	**88.0**	**92.6**	85.3	**92.6**	60
ours	93.4	95.9	86.0	90.7	**87.2**	91.6	63.4

TABLE IV

DETAILED F_1 AND IOU VALUES OF PROPOSED MODEL ON TWO DATA SETS

dataset	Im.Surf.		Build.		L.Veg.		Tree.		Car.		summary			
	IoU	F_1	IoU	F_1	IoU	F_1	IoU	F_1	IoU	F_1	mIoU	mF_1	AA	OA
Potsdam	89.1	94.2	94.5	97.2	80.5	89.2	80.9	89.4	90.7	95.1	87.1	93.0	91.5	92.2
Vaihingen	87.6	93.4	92.1	95.9	75.4	86.0	82.9	90.7	77.3	87.2	83.1	90.6	91.8	91.6

performance of other models. However, as Table II shows, compared with [27], our F_1 score on building is lower. But note that on the tree category, we exceed it. This may be explained that since tree and low vegetation usually coexist, there are two categories that are almost indistinguishable from the confusion matrix. So, their model may focus more on relatively large categories, such as building and low vegetation, while our model is more focused on small targets, such as trees or cars. This can also explain that we surpass them by 0.5% on car category. The visual comparison with ReSegNet is shown in Fig. 7.

E. Results on Vaihingen

Since the image sizes of the Vaihingen data set are irregular, we first expand the image and then crop image patches with a size of 512 × 512 pixels in an overlapping manner. Apart from methods mentioned above, there are many other works that use this data set, parts of which are briefly described below.

1) ADL: This model is proposed by Paisitkriangkrai *et al.* [50] in 2015. Their model can be divided into three parts: feature extraction, classifier, and post-processing. In the feature extraction part, CNN and hand-craft features are extracted. As for the classifier, a fully connected layer and random forest are utilized for CNN and hand-craft features, respectively. After that, the conditional random field is utilized as post-processing. Analogously, our model can be divided into feature extraction and classifier, but there are no hand-craft features in the feature extraction and our classifier is designed on the basis of neural network. In addition, there is no post-processing procedure and extra DSM data utilized in our model.

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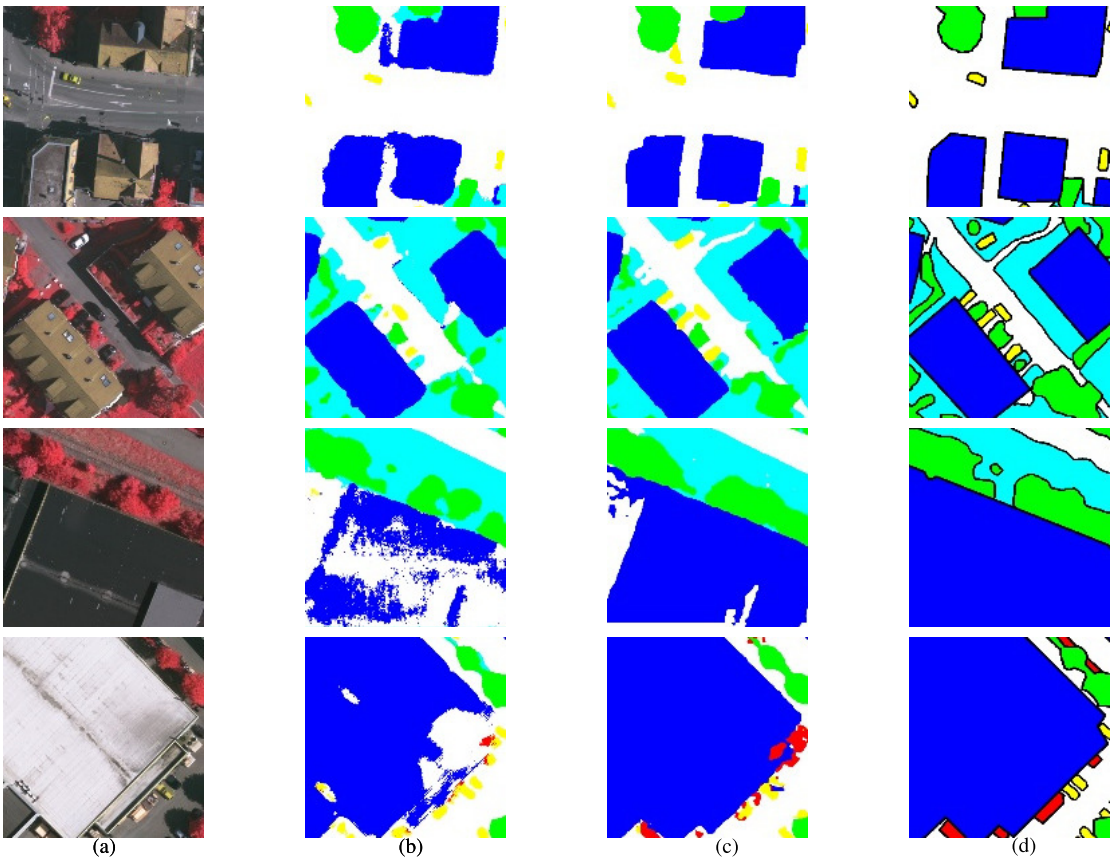


Fig. 8. Partial visualizations of results on Vaihingen data set. From left to right are raw image, predictions by ReSegNet and our AFNet, and the ground truth label, respectively. The colors on ground truth are, white: impervious surface, yellow: car, red: background, green: tree, cyan: low vegetation, blue: building. (a) Image. (b) ReSegNet. (c) Ours. (d) G.T.

2) DLR_10: This model is proposed by Marmanis *et al.* [52] in 2018. In order to alleviate the problem of feature maps with large receptive fields losing high-frequency information and causing blurred boundaries, they construct a comparatively simple model by adding boundary detection to existed multilevel architectures, and the boundary extraction network precedes the feature extraction network. In general, we both utilize multilevel structure, but their multilevel structure is composed of three independent subnetworks. As for the problems introduced by the large receptive field, we use SLAM and SFAM to solve rather than introducing an additional boundary detection network.

3) FuseNet: This model is proposed by Bergado *et al.* [51] in 2019. ReuseNet is proposed to solve the high inter-class spectral similarities, mismatching resolution of available bands, and regularization of noisy predictions. It is built on top of FuseNet by incorporating recurrent connections. FuseNet is a typical encoder–decoder structure architecture which takes panchromatic images and corresponding multispectral images as inputs. Then the score map produced by FuseNet, together with original panchromatic and multispectral images are feed into the next FuseNet to reproduce score map. In general, we both try to improve recognition ability and get better predictions. To this end, we both use multilevel structure and get the score map in a recurrent manner. However, our ACSR is more like a classifier, while they gradually refine the score map.

Table III shows the quantitative comparison with current SOTA methods on the ISPRS Vaihingen test set. Since Vaihingen with many detached buildings, the difficulty of building classification task has increased. Moreover, on the classification of vehicles, accuracy is significantly lower compared with the Potsdam data set. The phenomenon of reduced accuracy also appears in other work, such as in [48] and [49]. This is because the number of vehicles available for training in the Vaihingen data set is small. In general, our result is competitive with ReSegNet [27], and our performance of vehicles is much more accurate than them, just as Fig. 8 shows.

V. Discussion

A. Parameter Scales and Model Complexity

The parameter scales, floating point operations (FLOPs), and inference time are provided for further evaluation of the proposed method. The parameter scales are shown in Tables II and III. The values of comparison methods are provided in [46] or from the articles, and our parameter scale is calculated by Tensorflow [38]. From Tables II and III, we can see that compared with other models, the parameters of our model are medium. Since the quality of the hardware (e.g., Center Processing Unit and Graphics Processing Unit) also affects the training time of the model, so running times seem not to be reasonable for the evaluation of time complexity. Here, FLOPs and inference time, which are used in deep learning, are

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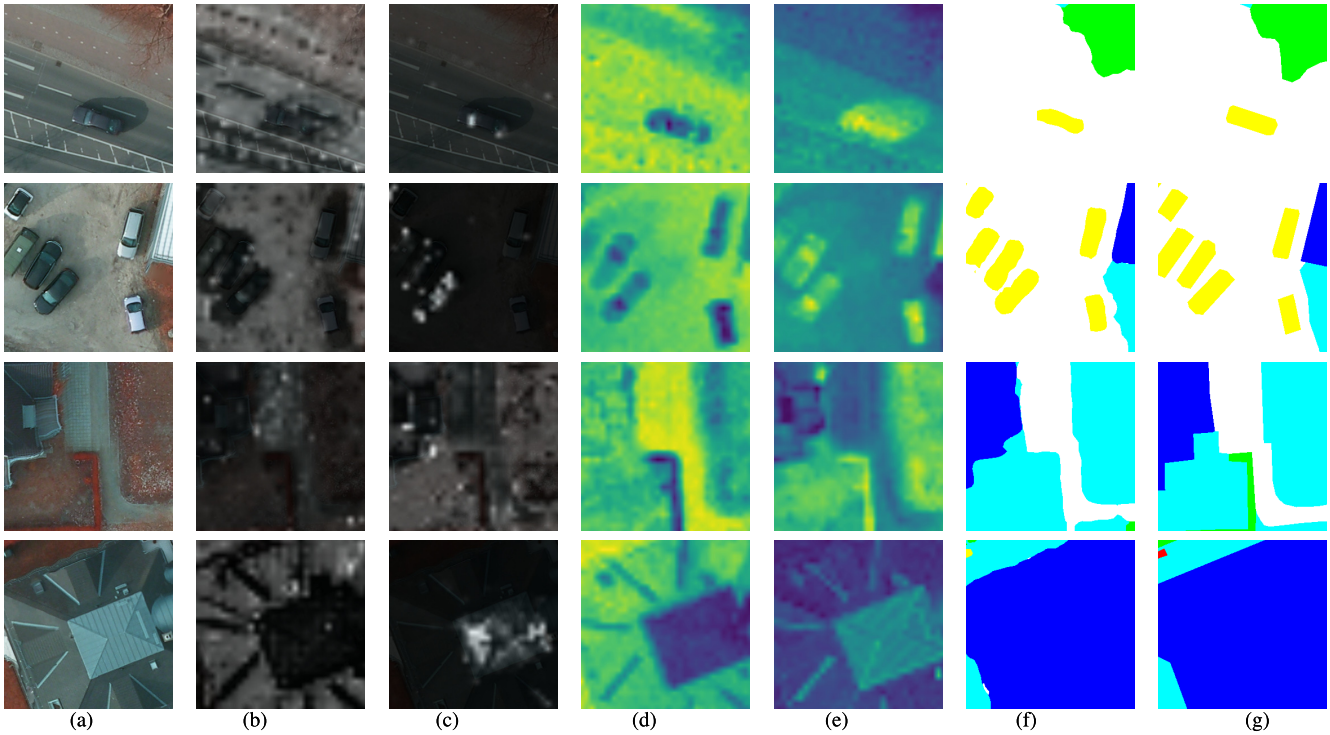


Fig. 9. Partial visualization of attention maps and feature maps. For each row, from left to right are the original image, attention maps, feature maps, prediction, and ground truth. On attention maps, the brighter area means more significant than other areas. On feature maps, the higher the color temperature is, the larger the feature value is. Because there may be multiple categories to be classified on the image, it is necessary to consider whether all the categories have been noticed, which is also the meaning of attention 1 and attention 2. The colors on ground truth are, white: impervious surface, yellow: car, red: background, green: tree, cyan: low vegetation, blue: building. (a) Image. (b) Attention1. (c) Attention2. (d) Feature 1. (e) Feature 2. (f) Prediction. (g) G.T.

TABLE V

QUANTITATIVE COMPARISONS (%) AMONG ABLATION STUDIES ABOUT PROPOSED MODULES ON ISPRS POTSDAM CHALLENGE VALIDATION SET. F IS SFAM, L IS SLAM, AND A IS ACSR MODULE. THE VALUES IN BOLD ARE THE BEST RESULT

Method	Im.Surf	Build.	L.Veg.	Tree	Car	OA	mF_1
deeplab	90.91	95.94	83.73	84.13	88.26	88.56	88.53
deeplab+F.	92.15	96.49	84.77	84.04	92.81	89.56	90.05
deeplab+L.	91.84	96.15	84.69	84.37	90.81	89.42	89.57
deeplab+F+L.	92.91	96.44	85.47	83.89	**93.74**	90.16	90.44
deeplab+F+L+A.	**93.16**	**96.77**	**86.68**	**85.24**	92.69	**90.68**	**90.91**

introduced to evaluate the complexity of our model. However, the FLOPs has not been listed in many other works, and inference time is relevant to the size of prediction patch, which vary from one to another. Compared with parameter scales, they are more susceptible to other factors. So only parameter scales are listed in Tables II and III. When training, the FLOPs of our model is about 4×10^{10}. When inference, it takes 0.075 s for AFNet to predict a semantic segmentation result for a sample averagely.

B. Model Analysis

In this part, we perform comprehensive ablations to validate each module of our model. As a baseline, we train deeplab-v3 [31] on infrared, red, green (IRRG) bands. Then we gradually add SFAM, SLAM, and ACSR. We conduct experiments on those architectures and report the performance via OA, mean F_1 $score$.

Table V shows our experimental results, where F is the abbreviation of SFAM, L is SLAM, and A is ACSR. We can see that the deeplab model is competitive with the others, which shows its capability of capturing contextual information for remote sensing image patches. As shown in the second and third rows of Table V, the OA increased by 1% and 0.86% when adding SFAM and SLAM to the deeplab, respectively. It indicates that when adding SLAM and SFAM independently to the deeplab, there is limited performance improvement. But when combining SLAM and SFAM together, the OA and F1 increased by 1.6% and 1.91%, respectively, which proves our hypothesis to some extent.

In addition, the classification accuracy for each category has been improved. Compared to other categories, as illustrated in Fig. 1, the semantic information of car is easily covered by other categories. When applying SFAM to the baseline, the performance of most categories has been improved, especially the performance of the car category. This indicates that the network can better solve the labeling of size-varied objects. Still, when applying SLAM to tackle the receptive field imbalance problem, the classification accuracy has been improved a lot.

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TABLE VI

ABLATION STUDY ON THE PREDEFINED $\bar{N}$. EXPERIMENTS ARE CONDUCTED ON THE VALIDATION SET. THE $\bar{N}$ IS SET TO 64, 128, 256, 512, RESPECTIVELY. ALL OTHER CONFIGURATIONS KEEP THE SAME. THE RESULTS ARE PRESENTED AS PERCENTAGES

$\bar{N}$	Im.Surf.		Build.		L.Veg.		Tree.		Car.		summary				# Para $\times 10^6$
	IoU	F_1	IoU	F_1	IoU	F_1	IoU	F_1	IoU	F_1	mIoU	mF_1	AA	OA	
64	90.43	92.26	95.09	96.52	70.22	85.51	67.96	84.71	91.38	93.63	83.02	90.53	87.06	89.79	49.99
128	90.49	92.79	95.02	96.55	73.48	86.02	67.61	85.56	92.12	95.42	83.74	91.27	88.19	90.30	54.04
256	91.36	93.16	94.87	96.77	76.71	86.68	69.16	85.24	92.57	92.69	84.93	90.91	88.24	90.68	63.40
512	91.39	93.19	94.83	96.79	77.39	86.86	69.55	85.39	92.41	92.79	85.11	91.00	87.82	90.71	80.41

Besides, as the last row of Table V shows, when the ACSR is added to the network, the classification accuracy of most categories has improved, which shows the need of a powerful classifier for this challenging task.

We visualize the attention maps generated by the network in Fig. 9 to better understand the effect of the attention module. Since high level features are too abstract to understand, here we visualize the attention maps and feature maps from the first SFAM and then upsample them by four times to recover the spatial resolution. The attention map is superimposed with the original image to help us understand the meaning of attention. The brighter the region, the greater the attention value. Generally, a remote sensing image comprises a complex landscape, there is more than one category to be classified on the image (e.g., as the ground truth of Fig. 9 shows). Just like the human perception process, for the attention mechanism, different categories are perceived asynchronously. So, different categories should have different maps. As the second row of Fig. 9 indicates, attention 1 focuses more on the impervious surface while attention 2 focuses more on the car (especially one of the cars). In addition, even within categories, there will be huge differences in characteristics. We humans will observe the various regions with huge differences, same as the attention mechanism. As the fourth row of Fig. 9 shows, the first attention map focuses more on the center region of the roof, the second attention map perceives other regions of the roof. In summary, different attention maps perceive different regions on the image, these regions may belong to different categories, or belong to the same category, but with large differences. When the attention maps are applied on the feature maps, it is clear that the features become more discriminative (see feature maps in Fig. 9).

C. Study of Free Parameter

Among all the free parameters, $\bar{N}$ could be the most significant one. From III-B, we can see that the predefined $\bar{N}$ determines the dimension of the embedding space and the parameter scale of the proposed model. So, in this section, we also discuss the influence of $\bar{N}$. Specifically, we set $\bar{N}$ to 64, 128, 256, 512, and keep other configurations consistent, explore the effect of $\bar{N}$ on the model. The results are presented in Table VI.

From Table VI, it is obvious that OA and mIoU increase monotonically with the predefined $\bar{N}$ in evaluation, and they seem to have a tendency to converge. This is reasonable, the larger $\bar{N}$ is, the greater the fitting ability of the model is since $\bar{N}$ introduces more trainable parameters. However, more parameters will lead to more calculations, longer training times, and bring challenges to optimization. It is necessary to find a tradeoff between parameter scale and performance.

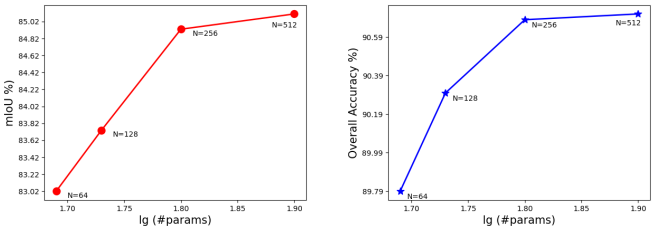


Fig. 10. Comparison between different $\bar{N}$, in terms of the parameter scale and OA, mIoU. For convenience, the x-axis of both subfigure is logarithmic. Those two subfigures show an upward tendency, and as the amount of parameters increases, the upward trend gradually slows down.

Fig. 10 shows the changes in OA, mIoU, and parameter scale when $\bar{N}$ values are different. Generally, the most idea model should have fewer parameters and higher accuracy, which is in the upper left corner area of Fig. 10. However, the actual situation is often to find a tradeoff between parameter scale and accuracy, which is in the upper-middle corner of Fig. 10. Our results show that as $\bar{N}$ increases, there is an upward tendency of both OA and mIoU. When $\bar{N}$ gets higher, the upward trend has slowed. This indicates that when $\bar{N}$ is set to 256, the parameter scale and accuracy of the model can reach a relative balance.

D. Input Size Matters

When we verify the robustness of the model on the validation data set, the size of the image patch is 300, the four sides (or two) of a small image patch can be named as borderlines. After merging the small patch into tiles with none overlapping manner, we can get the prediction map. By comparing the prediction map and ground truth, it can be found that there will be some unnatural horizontal or vertical lines on the prediction map. These lines are not the boundaries of objects. Moreover, the intersection of these unnatural lines and the predefined borderlines is the location where the border effect appears. Actually, we observe an obvious border effect in some areas of the prediction map. This effect results in many wrong

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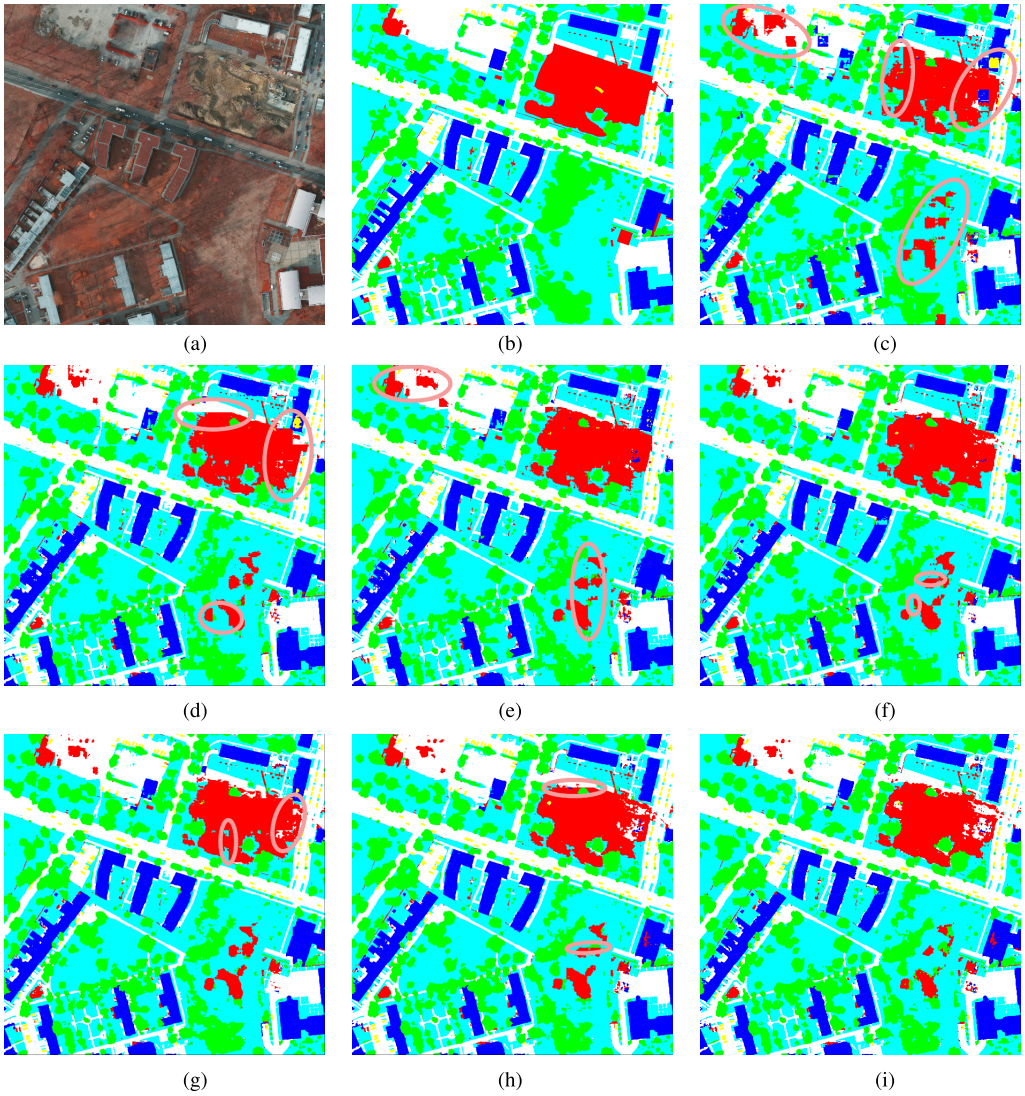


Fig. 11. Partial visualizations of the results obtained from different size inputs. The areas with obvious border effect are circled with pink ellipses. (a) Image. (b) Ground Truth. (c) Size 200. (d) Size 300. (e) Size 400. (f) Size 500. (g) Size 600. (h) Size 1000. (i) Size 3000.

TABLE VII

QUANTITATIVE COMPARISONS (%) AMONG ABLATION STUDIES ABOUT THE SIZE OF INPUT PATCH ON ISPRS POTSDAM CHALLENGE VALIDATION SET. THE VALUES IN BOLD ARE THE BEST RESULT

	200	300	400	500	600	1000	3000
OA	89.69	90.68	90.65	**91.0**	90.83	90.69	90.64
mF_1	86.43	87.77	87.0	**88.32**	88.07	87.55	87.73

classified pixels which declining the performance. A nature idea is to enlarge the input size of the image patch. When it has the same size as the original tile, the border effect would disappear. However, limited by GPU memory, the solution is not feasible. Thus, we crop patches with different sizes to search for the optimal patch size. A series of experiments on input size is shown below.

As Table VII shows, when the input size is 500, both OA and average F_1 reach the maximum. We can see that OA and F_1 score show an upward trend in the range of 200–500. After 500, the accuracy will decrease, but the decline is not obvious. This phenomenon may imply that when image size is smaller than 500, the accuracy is mainly influenced by the border effect. But with the image size increased, the border effect is gradually reduced, the accuracy is mainly influenced by the image size. The larger the image, the more difficult samples will be on the image, and the scale problem will be more severe which will degrade the performance of our model. We visualized the predictions to support our hypothesis.

In Fig. 11, we circle some obvious areas affected by the border effect. As Fig. 11 shows, the smaller the patch size, the more severe the border effect, which will reduce the performance of our model. The larger the patch size, the border effect is less obvious. For example, in Fig. 11(i), there are four borderlines in the middle of the image tile, however, these four lines do not show unnatural misclassified part. In addition, the area where the border effect appears is irregular, which may be the reason for the irregularity of the accuracy of small input images. Moreover, considering the confusing area in the top right corner of the image, as the size of the input image increased, the accuracy of this area appears to be significantly degraded.

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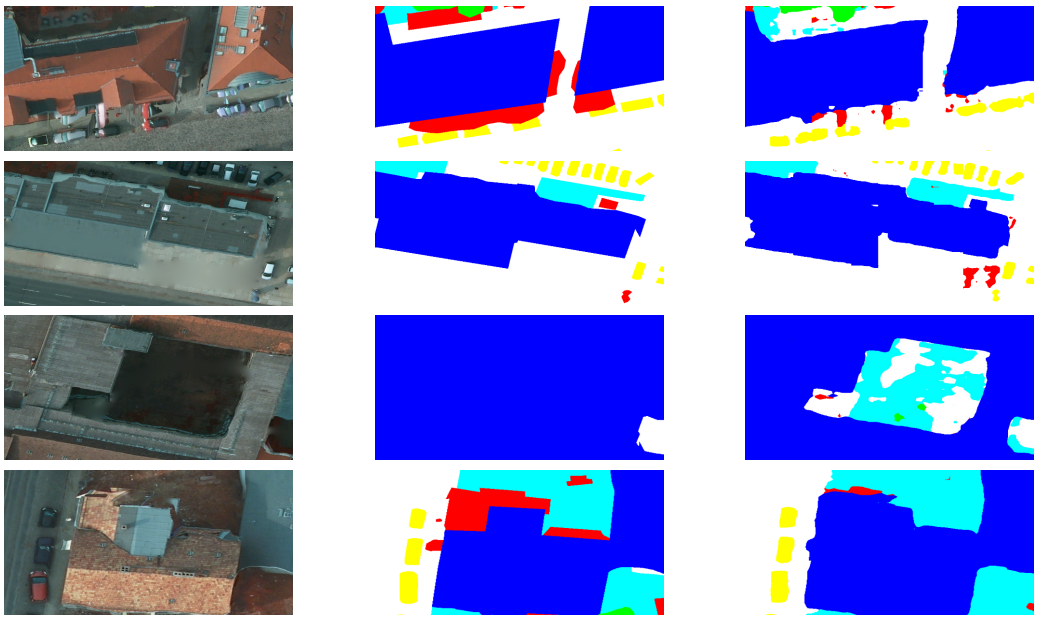


Fig. 12. Partial visualizations of the results obtained from uncertain areas. From left to right are image, prediction and ground truth. There are quality issues, e.g., mosaic, image distortion, incorrect labeling) either on image or ground truth, we display our prediction under these conditions. The colors on ground truth are, white: impervious surface, yellow: car, red: background, green: tree, cyan: low vegetation, blue: building. (a) Image. (b) Label. (c) Prediction.

E. Robustness to Uncertainties

The robustness of the model is discussed under two conditions. One is when the image quality is poor, and the other is when the label quality is poor. Due to sensor failure or other reasons, there are some regions with poor image qualities on the optical images. However, on the groundtruth image, these regions are sometimes not labeled as background but as its true category, or just the opposite, they are labeled as background. These uncertainties caused by image quality will bring an additional challenge to the labeling task, for those regions usually share quite different features. However, under this circumstance, our model can correctly predict the category of the area with poor imaging quality. For instance, in the first row of Fig. 12, the imaging of the vehicle is deformed. The deformed parts of cars appear on the impervious surface, which makes the impervious surface labeled as background on the ground truth image. But as shown in the prediction, most deformed parts are correctly inferred as impervious surfaces. In the second row of Fig. 12, there is a huge region covered with mosaic, and the region is not labeled as background but impervious surface (which is consistent with the results of interpreters). while our network can still correctly infer the region. This shows that our network can make inferences based on deep semantic features, so as to obtain an output closer to the true category of the region, even under the poor image quality condition.

Besides, the semantic labels in the ground truth may also be ambiguous. The given ground truth images may not reflect the true images. As shown in the third row of Fig. 12, there is an area covered with vegetation in the middle of the house. But this area is not labeled as vegetation but as a building. Similarly, in the fourth row of Fig. 12, the image of the house is partially tilted, and this area is labeled as the background instead of the building. These uncertainties caused by labels also bring challenges to the labeling task, which expands the feature space of the specific category. However, our network tends to predict the true category of a pixel, even if the predicted category does not match the labeled category.

TABLE VIII

QUANTITATIVE RESULTS (%) OF PERFORMANCE IMPROVEMENT BROUGHT BY DIFFERENT DATA ENHANCEMENT METHODS. ORIGIN MEANS THAT THE TEST IMAGE DOES NOT USE ANY DATA AUGMENTATION METHODS, R MEANS ROTATION AND FLIPPING, AND O MEANS THE TEST PATCH IS CROPPED IN AN OVERLAPPING MANNER. THE SIZE OF INPUT PATCHES IS ILLUSTRATED IN THE EXPERIMENT PART

method	Potsdam				Vaihingen			
	mIoU	mF_1	AA	OA	mIoU	mF_1	AA	OA
origin	83.9	91.1	89.4	90.8	80.9	89.3	89.9	90.7
origin+R	86.3	92.5	91.0	91.7	82.3	90.1	91.1	91.3
origin+O	84.3	91.4	90.0	90.9	81.4	89.5	90.3	90.9
origin+R+O	87.1	93.0	91.5	92.2	83.1	90.6	91.8	91.6

F. Data Augmentation on Test Set

Apart from the size of the input patch, the data augmentation could also influence the final performance. As is mentioned above, when in the test phase, we rotate the input patch by three angles, flip the input patch twice, and crop the patch in an overlapping manner for the final prediction. In this part, to evaluate the performance brought by the aforementioned data augmentation methods, we successively adopt the strategy of rotation and overlapping cropping and observe the performance improvement. The evaluation results are listed in Table VIII.

From Table VIII, we can see that for both data sets, rotation and flipping data augmentation methods can significantly improve the performance of the model. This may be because

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when the image is in different directions, the attention map is different for the same object. When combining attention maps from all directions, the attention map will perceive a more large area rather than a single local area of the object. In addition, when we crop the patches in an overlapping manner, the performance improvement on both data sets is limited. This is reasonable since the overlap sampling strategy is used to improve the classification accuracy of the border area of the image, especially when the size of the input patch is small. However, We crop the image patch in a larger size. Therefore, the border effect is not so great for the accuracy of the prediction of the patch.

VI. Conclusion

In this article, to improve VHR remote sensing image segmentation's performance, which is limited by size-varied and easily confused objects, we propose a novel AFNet. It consists of three modules, i.e., SFAM, SLAM, and ACSR. By SFAM, AFNet could adaptively enhance feature points from either low-level or high level feature maps, based on the categories of the points on the original image. By SLAM, for the multiscale feature map fusion, AFNet could adaptively enhance the feature maps whose receptive field matches the scale of the easily confused object. Furthermore, the ACSR makes AFNet more competitive in classifier design. We prove the effectiveness of the proposed module by extensive experiments and achieve the SOTA performance. However, since there are multisource remote sensing data (e.g., DSM), our proposed module can be further adapted to fuse feature maps from optical image and DSM, which will improve the performance of semantic labeling theoretically. That is also our future direction.

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